HW 4 · Learn a Score and Sample with a Probability-Flow ODE

Programming Assignment · Mathematical Foundations of Diffusion Models · Monsoon 2026 · IIT Jammu

Release date: TBA **Due:** Tuesday, 10 November 2026 **Format:** individual programming assignment; submit one executed Jupyter notebook (`.ipynb`) **Marks:** 36 points

Problem setup

Use one-dimensional data from the standard Laplace distribution

$$p_0(x) = \frac{1}{2}e^{-|x|}, \qquad x \in \mathbb{R}.$$

Compare these two signal schedules on $0 \le t \le 1$:

$$\alpha_t^{\text{lin}} = 1 - t, \qquad \alpha_t^{\text{exp}} = e^{-\beta t}, \quad \beta = 8.$$

For either schedule, form forward-corrupted data as

$$X_t = \alpha_t X_0 + \sqrt{1 - \alpha_t^2}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1).$$

The linear schedule reaches a standard Gaussian exactly at $t = 1$. The exponential schedule retains a small data contribution at $t = 1$.

Problem 1. Generate and inspect the forward data — 8 points

1. Simulate independent training, validation, and test data from p_0.
2. For both schedules, generate noisy training examples at randomly sampled times.
3. Plot the clean data and forward-corrupted distributions at several times. Compare the terminal distributions with a standard Gaussian.

Problem 2. Train score-estimation networks — 8 points

For each schedule, train a neural network using forward-corrupted training data. Compare:

1. a network that receives raw time t;
2. a network that receives t with Fourier positional encodings.

Use held-out data to compare the models. Include the training setup and a plot of the training and validation losses.

Optional 2(b). Inspect the estimated score — ungraded

Plot the estimated marginal score at selected times and compare it with a numerical reference from the known Laplace distribution.

Problem 3. Sample with the probability-flow ODE — 12 points

For each trained model, construct the probability-flow vector field for its forward schedule. Initialize samples from a standard Gaussian and integrate the ODE backward with an explicit Euler solver. Compare at least two solver resolutions.

Problem 4. Compare generated and target samples — 8 points

Compare the generated samples with the standard Laplace target using density plots and numerical summaries. Discuss the effect of the schedule, time encoding, and ODE resolution.

Submission

Submit one executed Jupyter notebook (`.ipynb`). Write the theoretical parts in Jupyter Markdown cells using LaTeX, and place them before the code that implements each part. Include the code, plots, results, and a brief explanation of what the results show. The notebook should run on a laptop CPU.